HAWLER MEDICAL UNIVERSITY

Introduction to Clinical Biochemistry

Spectrophotometer & ELISA

College of Medicine
Basic Science Department · 2nd Year Medical Students 2026

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Learning Outcomes

Define clinical biochemistry and explain its role in patient care.



Describe the measurement principle and main components of a spectrophotometer.



Apply the Beer–Lambert law to relate absorbance and concentration.



Explain the principle, formats, and clinical applications of ELISA.

Clinical biochemistry turns samples into decisions

Clinical biochemistry is the division of laboratory medicine that deals with the measurement of chemicals in **blood, urine, and other body fluids**.

Detection

and diseases early.
Identifying health problems

Prognosis

Determining the likely course and outcome of a disease.

Therapy

Guiding and monitoring patient treatment plans.

Spectrophotometry measures what a sample absorbs

The Core Principle

Spectrophotometry determines how much light a solution absorbs or transmits at a selected wavelength.

Wavelength Selectivity

Different chemical compounds absorb light at specific, characteristic wavelengths.

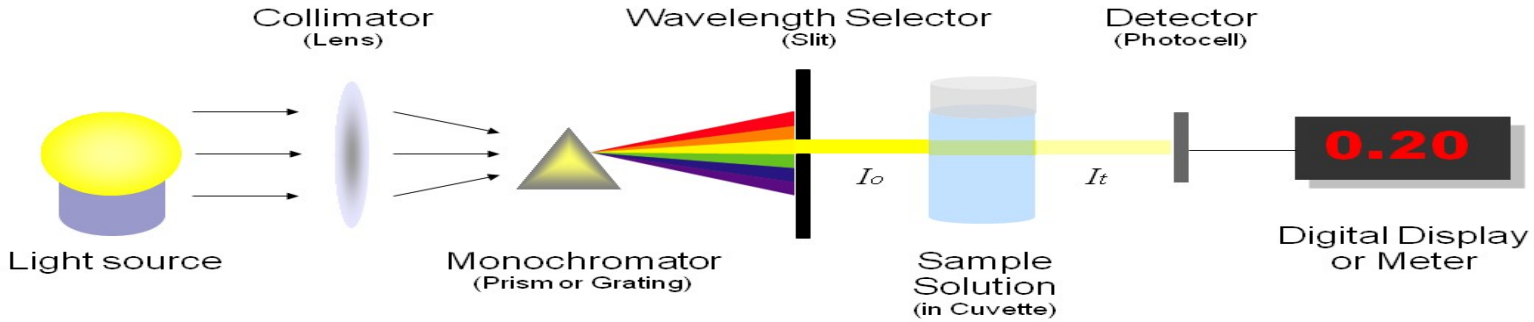
Quantitative Analysis

By measuring the light absorbed after passing through a cuvette, we can determine the concentration of an analyte. [2]





Components of Spectrophotometer



Component	Function & Role
Monochromato	Uses a prism or grating to disperse light into its spectrum.
Wavelength Selector	A slit that isolates the specific wavelength required for the assay.
Cuvett	A sample holder with a precise optical path length (l).
Photomete r	The detection system that quantifies transmitted light intensity.

Beer–Lambert law links absorbance to concentration

$$A = \varepsilon \cdot c \cdot l$$

A = Absorbance

A dimensionless measure of the light absorbed by the sample.

C = Concentration

The amount of solute in the solution (typically in mol/L).

ε = Molar Absorptivity

A constant specific to the substance and wavelength ($L \cdot mol^{-1} \cdot cm^{-1}$).

l = Path Length

The distance light travels through the sample (usually 1 cm).

At a fixed path length and wavelength, absorbance increases linearly with concentration.

[1]

Quantification compares the sample with a standard

THE RATIO PRINCIPLE

$$A_{st} / A_t = C_{st} / C_t$$

When the optical path length (L) is constant, the ratio of absorbances is directly proportional to the ratio of concentrations.

FINAL CALCULATION

$$C_t = (A_t / A_{st}) \times C_{st}$$

The unknown concentration (C_t) is determined by multiplying the standard concentration by the ratio of the test absorbance to the standard absorbance.

Good samples protect measurement accuracy

Sample Requirements

Optical Properties

Analyte must be colored or chemically converted to a colored product for detection.

Concentration Limits

Solutions must be dilute enough to stay within the linear working range of the law.

Physical Clarity

Samples must be clear and free of precipitates, bubbles, or turbidity that scatter light.

Laboratory Protocol

Use a clean, scratch-free cuvette for every measurement.
Verify the correct wavelength (λmax) for the specific analyte.
Always "blank" the instrument with a reference solution before reading.
Ensure the solution is homogeneous and free of air bubbles.

ELISA couples immune specificity to an optical signal

Enzyme-Linked Immunosorbent Assay (ELISA) is a biochemical technique that uses antibodies or antigens immobilized on a solid surface and an enzyme label to generate a measurable colorimetric signal. [3]

DIRECT

- Antigen is immobilized on the plate.
- Enzyme-labeled primary antibody binds directly.
- Fastest workflow; fewer steps.

INDIRECT

- Antigen is immobilized on the plate
- Primary antibody binds, followed by labeled secondary.
- Higher sensitivity and flexibility.

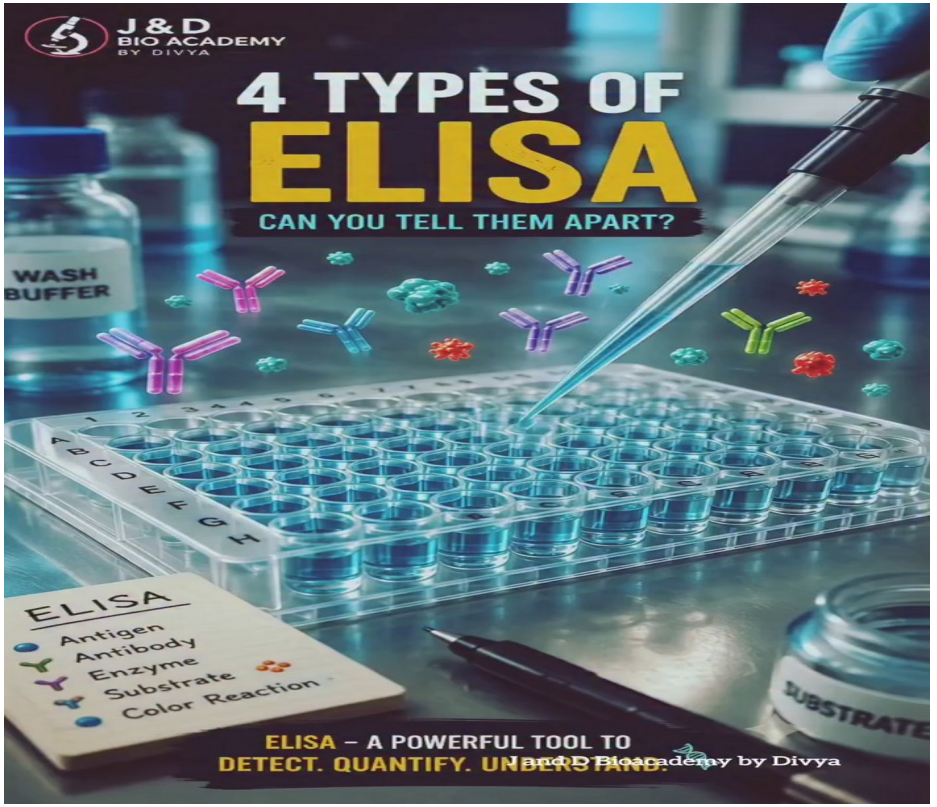
SANDWICH

- Capture antibody is immobilized.
- Antigen is "sandwiched" between two antibodies.
- High specificity for complex samples.

COMPETITIVE

- Sample antigen competes with labeled antigen.
- Signal is inversely proportional to concentration.
- Ideal for small molecules/hapten

Step-by-Step ELISA Machine Operation Guide



ELISA supports broad clinical laboratory applications

Hormones & Markers

- **Thyroid:** T3, T4, TSH
- **Reproductive:** HCG, LH, FSH
- **Metabolic:** Insulin, Cortisol
- **Tumor:** AFP, PSA, CEA

Infectious Disease

- **Viral:** Hepatitis B, HIV, COVID
- **Bacterial:** Toxins, Brucella, Salmonella
- **Antibodies:** Antiviral titers
- **Screening:** Point-of-care test

Immune Markers

- **Autoantibodies:** Anti-DNA, RF
- **Allergy:** Specific IgE levels
- **Cytokines:** IL-6, TNF-α
- **Monitoring:** Therapeutic drug levels

References

[1] **Chemistry LibreTexts.** "The Beer-Lambert Law." Physical and Theoretical Chemistry Supplemental Modules. https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Supplemental_Modules_(Physical_and_Theoretical_Chemistry)/Spectroscopy/Electronic_Spectroscopy/Electronic_Spectroscopy_Basics/The_Beer-Lambert_Law

[2] **Chemistry LibreTexts.** "2.1.5: Spectrophotometry." Experimental Determination of Kinetics. https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Supplemental_Modules_(Physical_and_Theoretical_Chemistry)/Kinetics/02%3A_Reaction_Rates/2.01%3A_Experimental_Determination_of_Kinetics/2.1.05%3A_Spectrophotometry

[3] **Zubair M, Singh C, Farhana A.** "Enzyme-Linked Immunosorbent Assay (ELISA)." StatPearls [Internet], NCBI Bookshelf. https://www.ncbi.nlm.nih.gov/books/NBK555922/

[4] **Cox KL, et al.** "Immunoassay Methods." Assay Guidance Manual [Internet], NCBI Bookshelf. https://www.ncbi.nlm.nih.gov/books/NBK92434/

LECTURE CONCLUSION

Thank You

Questions & Discussion

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